

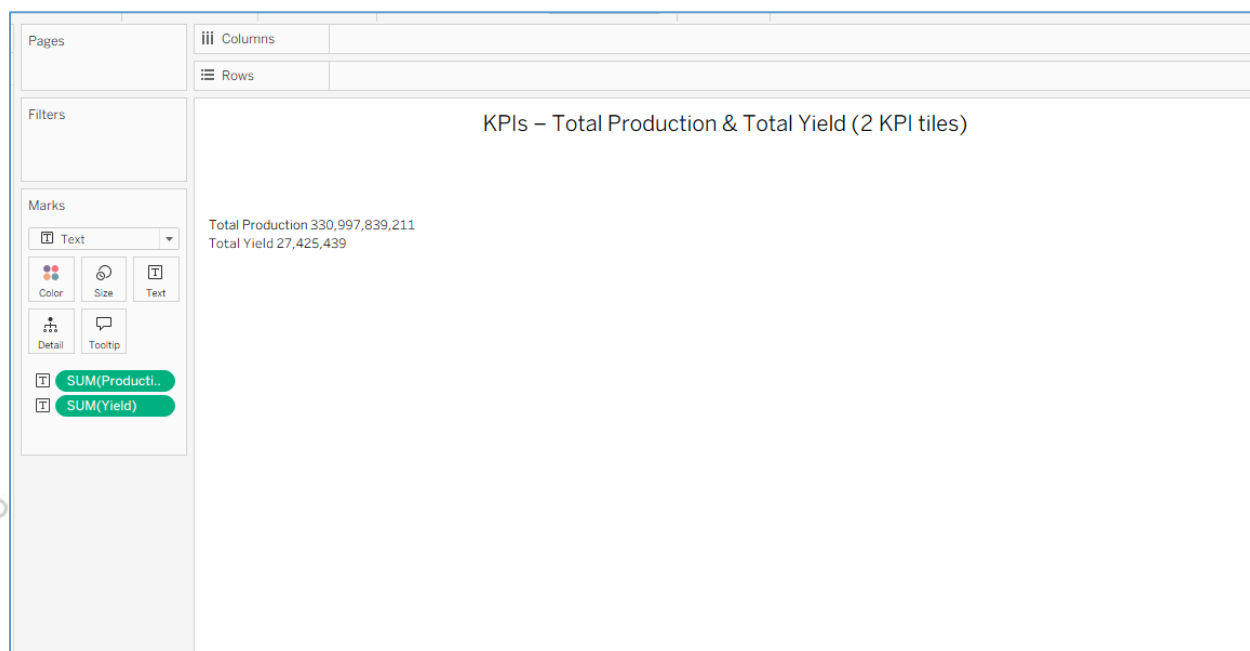
TEAM ID	2026-8827
PROJECT TITLE	India's Agricultural Crop Production Analysis
MAX MARKS	3 MARKS

6.3 – Data Preparation

Sheet – 01 – KPIs – Total Production & Total yield (2 KPI tiles)

Steps:

1. Open Tableau and connect to the agricultural dataset.
2. Verify fields such as Year, State, Crop, Season, Area, Production, and Yield are loaded correctly.
3. Create calculated fields for Total Production and Total Yield if required.
4. Create a KPI sheet and drag SUM(Production) to Text to display Total Production.
5. Drag SUM(Yield) to Text to display Total Yield in the same KPI sheet.
6. Format the KPI values with proper labels, font size, and alignment.
7. Create a line chart with Year on Columns and SUM(Yield) on Rows to analyze yield trends.
8. Create a state-wise production chart using State on Rows and SUM(Production) on Columns.
9. Create a crop-wise production chart using Crop and SUM(Production).
10. Build a state-wise cultivated area comparison chart using State and SUM(Area).
11. Create a Top/Leading States chart by sorting states based on Production or Yield.
12. Add filters for Year, State, Crop, and Season to enable interactive analysis.
13. Create a dashboard and place all KPI tiles and charts together.
14. Click on save.

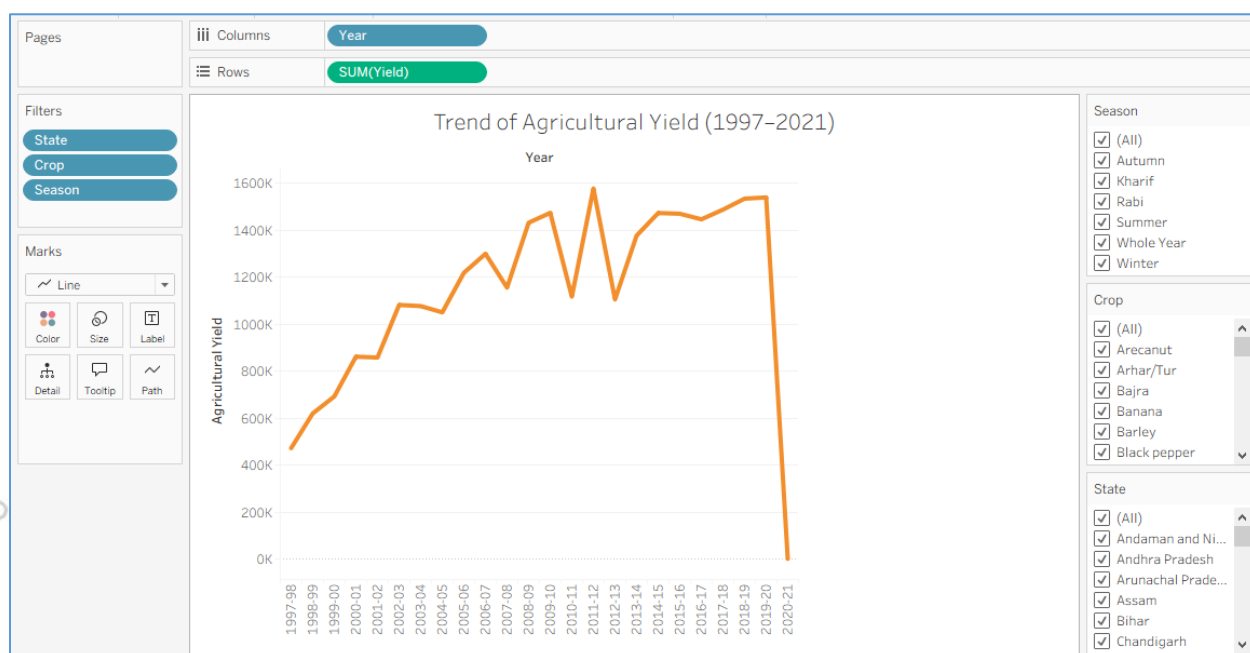


Sheet – 02 – Trend of Agricultural yield (1997-2021) – Line Chart

Steps:

1. Create a new worksheet and rename it Trend of Agricultural Yield (1997–2021).
2. Drag Year to the Columns shelf.
3. Drag Yield to the Rows shelf and keep aggregation as SUM.
4. Change the Marks type to Line to display the yield trend over time.
5. Sort the Year field in ascending order to maintain chronological sequence.
6. Add State, Crop, and Season to the Filters shelf.
7. Right-click each filter and select Show Filter to display filter cards.
8. Configure filters as Multiple Values (List) for interactive selection.
9. Format the Y-axis title as Agricultural Yield.
10. Add the worksheet title Trend of Agricultural Yield (1997–2021).
11. Customize line color, thickness, and font formatting for better visualization.
12. Verify the trend pattern and ensure filters update the chart dynamically.
13. Save the worksheet.

India's Agricultural Crop Production Analysis.



Sheet – 03 – State-wise Agricultural Production Distribution – India Map

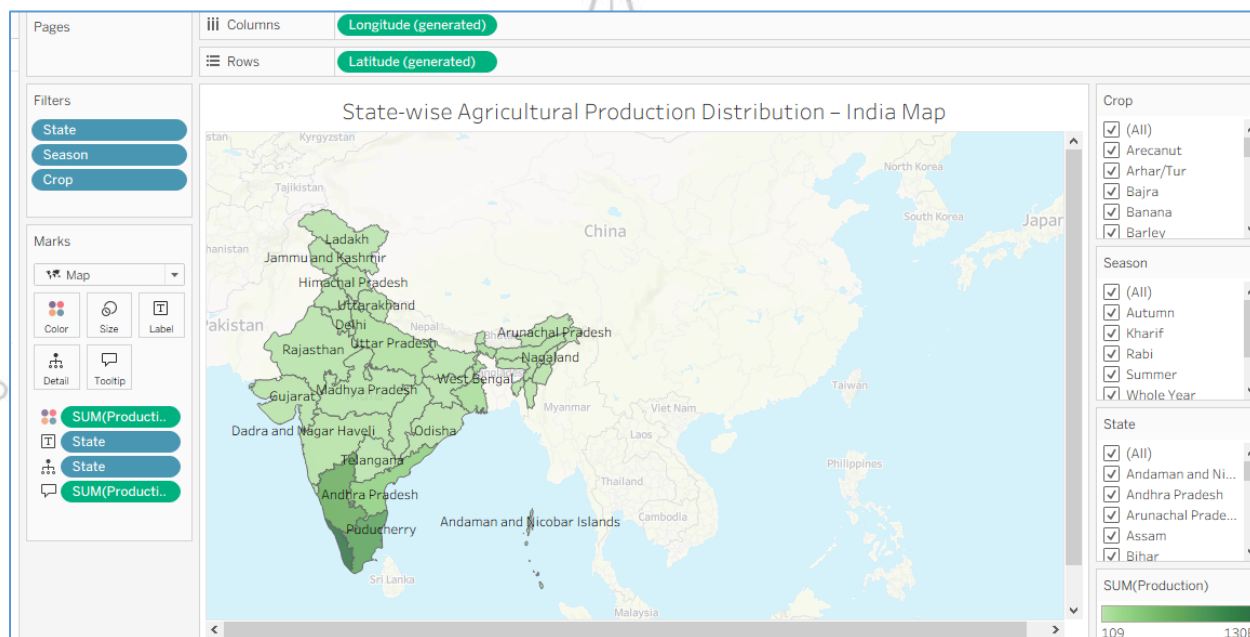
Steps:

1. Open Tableau and connect the Agriculture Production dataset.
2. Drag State to the workspace and set its Geographic Role as *State/Province*.
3. Double-click State to generate the India map.
4. Tableau automatically adds Longitude (generated) and Latitude (generated).
5. Change the Marks type to Map.
6. Drag Production to Color on the Marks card.
7. Drag Production to Tooltip for production details.
8. Drag State to Label to display state names on the map.
9. Drag State, Season, and Crop to the Filters shelf.
10. Show filter cards for State, Season, and Crop.
11. Select suitable color intensity based on SUM(Production).
12. Add the title “State-wise Agricultural Production Distribution – India Map”.

India's Agricultural Crop Production Analysis.

13. Format the map and labels for better visualization.

14. Save the worksheet.

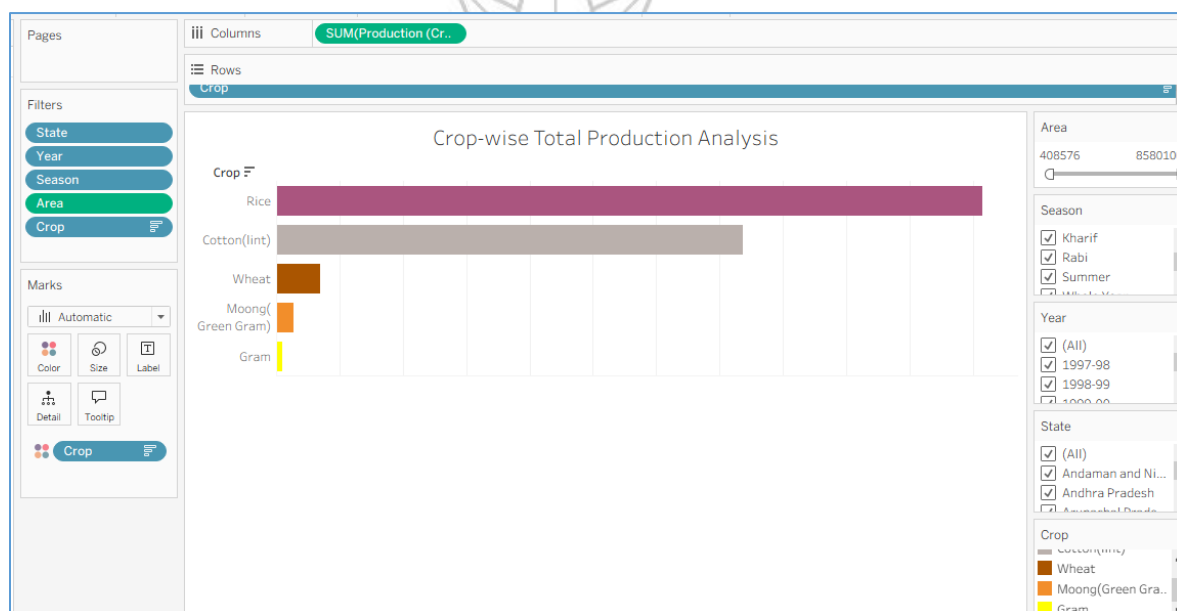


Sheet – 04 – Crop-Wise Total Production Analysis – Horizontal Bar Chart

Steps:

1. Open Tableau and connect the agricultural production dataset.
2. Create a new worksheet and rename it Crop-wise Total Production Analysis.
3. Drag Crop to the Rows shelf.
4. Drag Production (Crores) to the Columns shelf.
5. Select Bar Chart from the Show Me panel or Marks card.
6. Drag Crop to Color on the Marks card.
7. Drag Production (Crores) to Tooltip for detailed information.
8. Drag State, Year, Season, Area, and Crop to the Filters shelf.
9. Show all filter cards on the worksheet.
10. Sort the bars in descending order of production.
11. Adjust colors and formatting for better visualization.

12. Add the title "Crop-wise Total Production Analysis".
13. Verify that the chart displays total production for each crop.
14. Save the worksheet.



Sheet – 05 – State-Wise Cultivated Area Comparison – Grouped Bar Chart

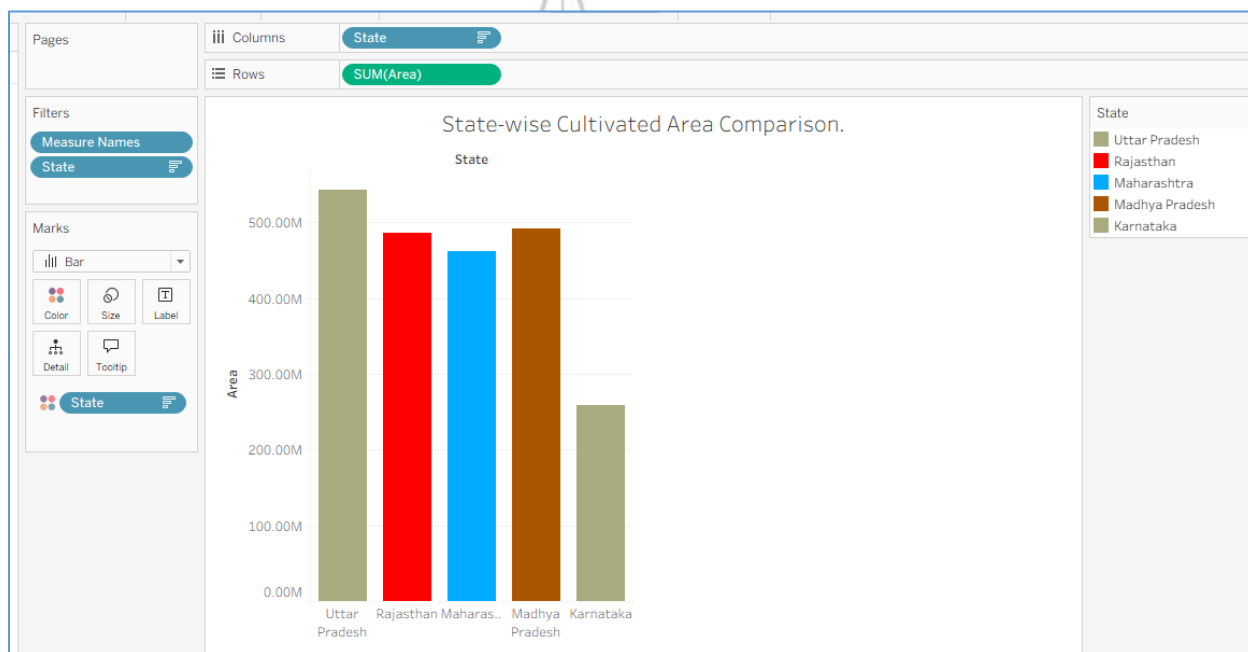
Steps:

1. Open Tableau and connect the agricultural dataset.
2. Create a new worksheet and rename it State-wise Cultivated Area Comparison.
3. Drag State to the Columns shelf.
4. Drag Area to the Rows shelf.
5. Set the Marks type to Bar Chart.
6. Drag State to Color on the Marks card.
7. Drag State to Tooltip for additional details.
8. Drag State and Measure Names to the Filters shelf.
9. Select the required states for comparison.
10. Format the chart colors and axis labels.
11. Add the title "State-wise Cultivated Area Comparison".
12. Verify that the bars represent the cultivated area of each state.

India's Agricultural Crop Production Analysis.

13. Adjust the chart layout for better readability.

14. Save the worksheet.



Sheet – 06 – Leading States by Agricultural Production – Bar Chart

Steps:

1. Open Tableau and connect the agricultural production dataset.
2. Create a new worksheet and rename it Leading States by Agricultural Production.
3. Drag State to the Columns shelf.
4. Drag Production (Crores) to the Rows shelf.
5. Select Bar Chart from the Marks card.
6. Drag State to Color to differentiate each state.
7. Drag State to Tooltip for state-wise details.
8. Drag State, Year, and Production (Crores) to the Filters shelf.
9. Apply filters to display the required states and years.
10. Sort the bars in descending order of production.
11. Format the axis labels and chart appearance.
12. Add the title "Leading States by Agricultural Production".
13. Save the worksheet.

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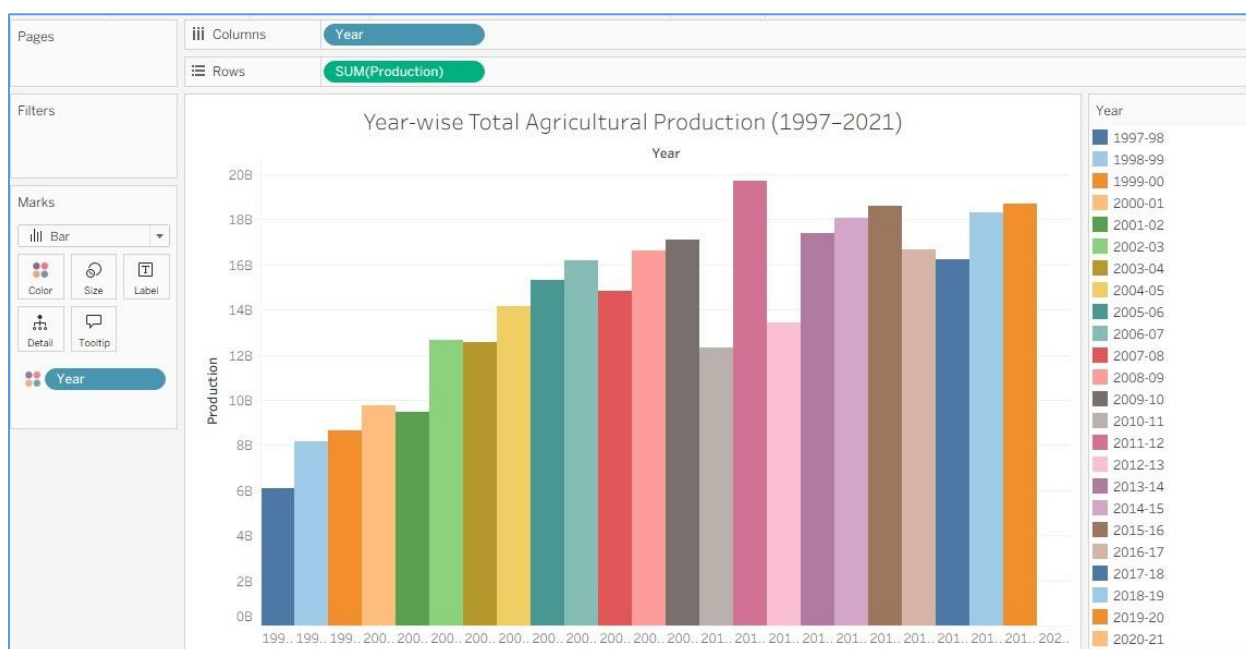


Sheet – 07 – Year-Wise Total Agricultural Production (1997-2021) – Bar Chart

Steps:

1. Open Tableau and connect the agricultural production dataset.
2. Create a new worksheet and rename it Year-wise Total Agricultural Production (1997–2021).
3. Drag Year to the Columns shelf.
4. Drag Production to the Rows shelf.
5. Select Bar Chart from the Marks card.
6. Drag Year to Color to display different colors for each year.
7. Drag Production to Tooltip for detailed values.
8. Format the Production axis and labels for better readability.
9. Sort the years in ascending order from 1997–98 to 2020–21.
10. Enable data labels if required.
11. Adjust the chart size and formatting.
12. Add the title "Year-wise Total Agricultural Production (1997–2021)".
13. Save the worksheet.

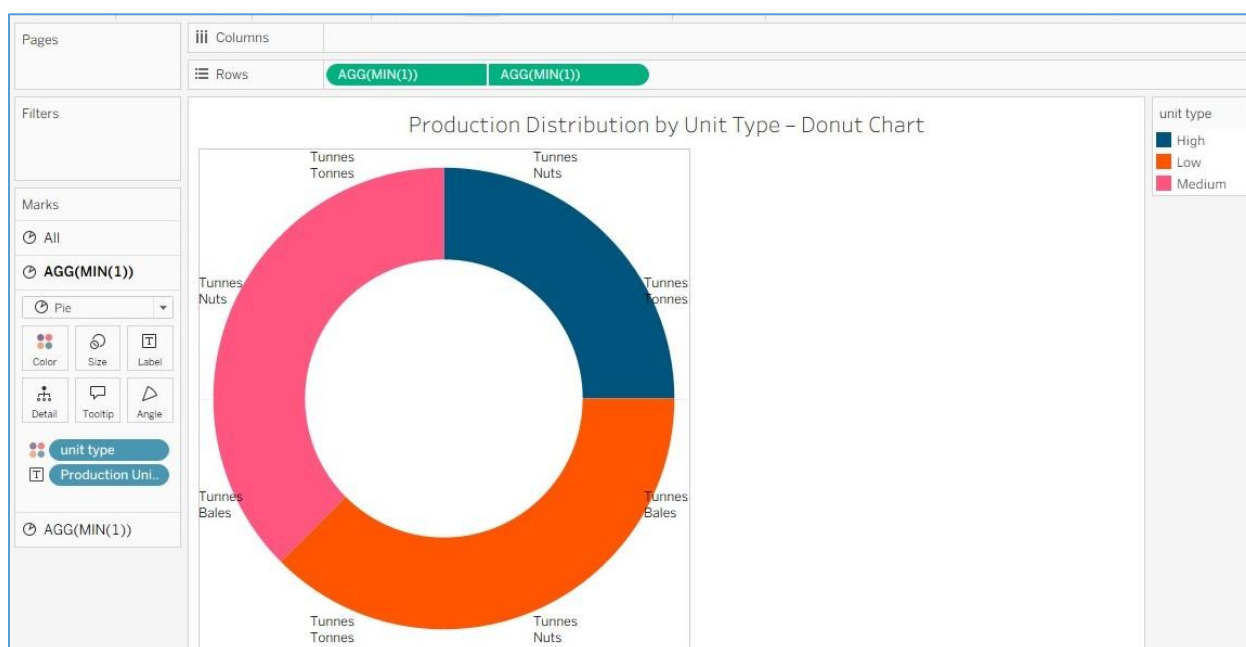
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Sheet – o8 – Production Distribution by Unit Type – Donut Chart

Steps:

1. Open Tableau and connect the agricultural production dataset.
2. Create a new worksheet and rename it Production Distribution by Unit Type – Donut Chart.
3. Create a calculated field MIN (1) and drag it twice to the Rows shelf.
4. Set the Marks type to Pie for both measures.
5. Drag unit type to Color on the Marks card.
6. Drag Production Units to Angle to define the pie slices.
7. Drag Production Units to Label to display values.
8. Right-click the second MIN (1) field and select Dual Axis.
9. Synchronize the axes and hide the headers.
10. Reduce the size of the second pie and set its color to white to create the donut hole.
11. Format colors, labels, and tooltips for better presentation.
12. Add the title "Production Distribution by Unit Type – Donut Chart".
13. Verify that each segment represents the distribution by unit type (High, Medium, Low)
14. Save the worksheet.

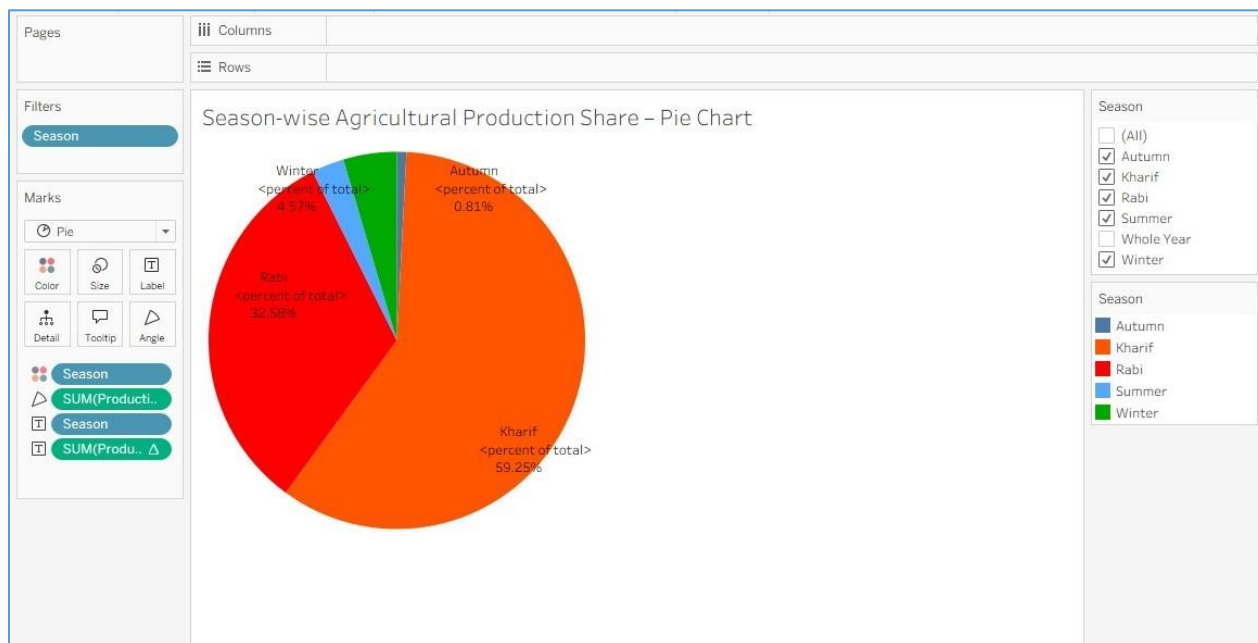


Sheet – 09 – Season-Wise Agricultural production Share – Pie Chart

Steps:

1. Connect your dataset containing "Season" and "Production" data.
2. Change the Mark type in the Marks card from "Automatic" to Pie.
3. Drag the "Season" dimension onto the Colour property.
4. Drag the "Production" measure onto the Angle property.
5. Drag "Production" again onto the Label property.
6. Right-click the Production label, select Quick Table Calculation, and choose
7. Percent of Total.
8. Drag "Season" onto the Label property to show the category names.
9. Drag "Season" onto the Filters shelf.
10. Save the worksheet.

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DELTA
– ACHIEVERS –